JAMES CARROLL

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PROFESSIONAL SUMMARY

I'm a full stack engineer with 12 years of experience building scalable web applications and backend systems. I've spent the last eight years at senior and staff level positions, leading technical decisions and mentoring development teams. My background spans early-stage startups through established tech companies, and I'm comfortable working across the entire stack from React frontends to distributed Python backends.

WORK EXPERIENCE

Staff Engineer

TechVault Solutions Ltd

techvault.io

London, United Kingdom

March 2021 - Present

Led the architecture and development of a microservices platform serving 2 million daily active users, reducing infrastructure costs by 34% through optimised containerisation and Kubernetes deployment strategies. Established engineering standards and code review practices across a team of 8 engineers, improving deployment safety and reducing production incidents by 48%. Designed and implemented a real-time data pipeline processing 500GB daily using Apache Kafka and Apache Spark, enabling the analytics team to reduce report generation time from 6 hours to 15 minutes. Mentored 4 junior engineers through promotion cycles and helped establish a technical mentoring programme across the engineering division.

Senior Full Stack Engineer

Nexus Digital Ltd

nexus-digital.co

Oxford, United Kingdom

June 2017 - February 2021

Rebuilt the core customer-facing web application using React and Node.js, improving page load times from 4.2 seconds to 1.1 seconds and increasing user retention by 22%. Implemented PostgreSQL database optimisation and query refactoring that reduced average API response times by 60% and eliminated database lock contention issues. Developed and deployed a customer analytics dashboard using D3.js and Python Flask that provided executives with real-time insights into user behaviour and product performance. Designed comprehensive API documentation and SDKs in Python, JavaScript, and Go, enabling third-party integrations that generated 18% of annual revenue.

Full Stack Engineer

Brightline Media Inc

www.brightlinemedia.com

Bristol, United Kingdom

September 2012 - May 2017

Built and maintained multiple Django and AngularJS applications serving advertising and publishing workflows for enterprise clients including national newspapers and digital media companies. Developed automated testing

frameworks and continuous integration pipelines using Jenkins and Docker, reducing deployment failures from 12% to 1.5% within the first year. Optimised MongoDB and Redis usage patterns across the application stack, improving performance during peak traffic periods and enabling the platform to scale from 50K to 500K daily requests without additional infrastructure. Participated in on-call rotation and resolved critical production incidents, maintaining 99.7% uptime across all customer-facing services.

EDUCATION

University of Oxford

MSc Computer Science

Specialisation in Distributed Systems and Software Architecture

Graduated: September 2012

SKILLS

Backend Development: Python, Node.js, Go, Java, Django, Flask, FastAPI, Express.js

Frontend Development: React, JavaScript, TypeScript, HTML5, CSS3, D3.js, Redux

Databases: PostgreSQL, MongoDB, Redis, Elasticsearch, Apache Cassandra

Cloud and DevOps: AWS (EC2, S3, Lambda, RDS), Docker, Kubernetes, Terraform, Jenkins, GitHub Actions

Architecture and Design: Microservices, REST APIs, GraphQL, Event-driven architecture, System design

Data Engineering: Apache Kafka, Apache Spark, Airflow, ETL pipelines

Testing and Quality: Jest, Pytest, Mocha, Test-driven development, Integration testing

Other: Linux, Git, SQL, UNIX shell scripting, Agile methodologies